

- 4 In thale cress, *Arabidopsis thaliana*, most of the stomata are on the lower surface of the leaves.

Fig. 4.1 is a diagram of an open stoma and two guard cells. Some of the cellulose fibres in the cell wall of the guard cells are shown.

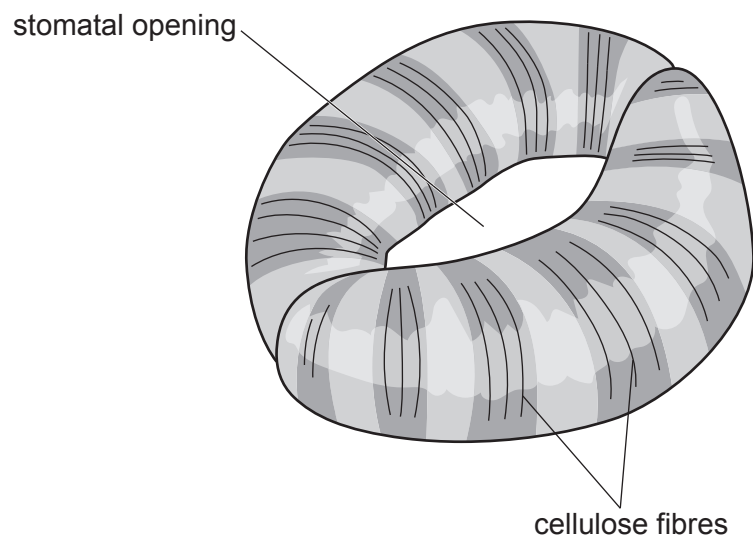


Fig. 4.1

- (a) The cellulose fibres shown in Fig. 4.1 are composed of bundles of cellulose microfibrils.

Explain how molecules of cellulose are arranged into a microfibril.

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(b) The width of stomatal openings in *A. thaliana* is regulated by movement of ions. These ions move through channel proteins in the cell surface membranes of the guard cells.

(i) Draw a diagram to show part of a cell surface membrane with a channel protein.

Label your diagram.

[3]

(ii) Explain why channel proteins are needed for the movement of ions into and out of cells.

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(c) The outward movement of ions from guard cells causes stomata to close.

A variety of *A. thaliana* does **not** have channel proteins for the outward movement of ions in the cell surface membranes of the guard cells.

Suggest **and** explain the effect of **not** having these channel proteins on transpiration.

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[Total: 10]